

Circuit Simulation Lab:18

Design of Counters

The purpose of this experiment is to introduce the design of Synchronous sequential circuits, in this case we are taking a 8 bit up counter.

CODE:

```
module up_counter_8bit(  
input clk,  
input rst,  
output reg [7:0] count  
);
```

```
always @(posedge clk)  
begin
```

```
if(rst)  
    count <= 8'd0;
```

```
else  
    count <= count + 1'b1;
```

```
end
```

```
Endmodule
```

Testbench

```
module tb_up_counter_8bit;
```

```
reg clk;  
reg rst;
```

```
wire [7:0] count;
```

```
up_counter_8bit dut(  
    .clk(clk),  
    .rst(rst),  
    .count(count)  
);
```

```
always #10 clk = ~clk;
```

```
initial begin
```

```
$dumpfile("counter8.vcd");  
$dumpvars(0,tb_up_counter_8bit);
```

```
clk = 0;  
rst = 1;
```

```
#20;  
rst = 0;
```

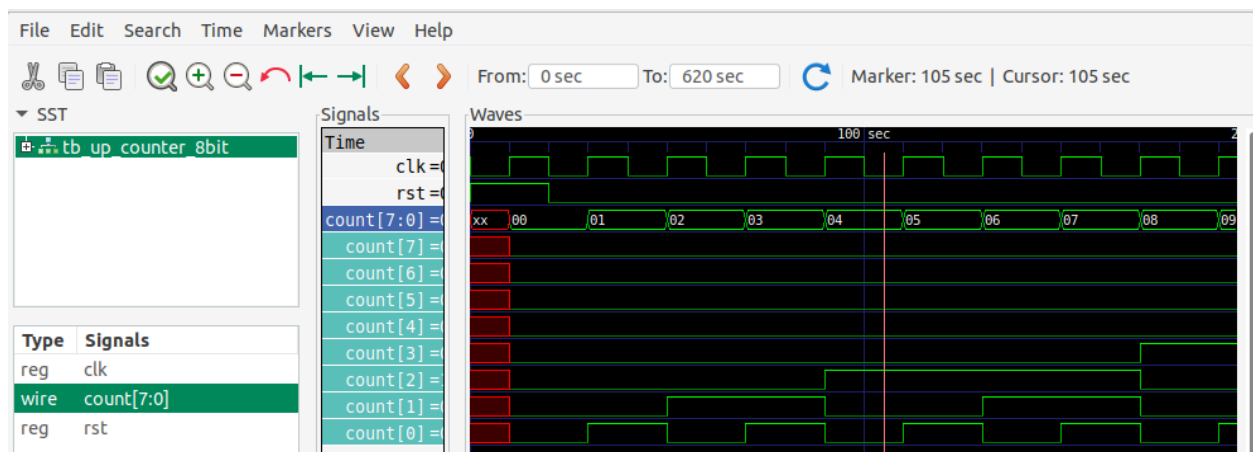
```
#600;
```

```
$finish;
```

```
end
```

```
initial begin  
$monitor("time=%0t rst=%b count=%d",  
        $time,rst,count);  
end
```

```
Endmodule
```



```
vlsi_dev_box@vlsi-box:~/Documents/NIELIT_ASSIGNMENTS/EXP_NO_18$ gedit tb_up_counter_8bit.v
vlsi_dev_box@vlsi-box:~/Documents/NIELIT_ASSIGNMENTS/EXP_NO_18$ iverilog -o out up_counter_8bit.v tb_up_counter_8bit.v
vlsi_dev_box@vlsi-box:~/Documents/NIELIT_ASSIGNMENTS/EXP_NO_18$ vvp out
VCD info: dumpfile counter8.vcd opened for output.
time=0 rst=1 count= x
time=10 rst=1 count= 0
time=20 rst=0 count= 0
time=30 rst=0 count= 1
time=50 rst=0 count= 2
time=70 rst=0 count= 3
time=90 rst=0 count= 4
time=110 rst=0 count= 5
time=130 rst=0 count= 6
time=150 rst=0 count= 7
time=170 rst=0 count= 8
time=190 rst=0 count= 9
```